

RESEARCH MASTERCLASS FOR CLINICIANS · MODULE 1

Introduction to Fundamentals of Epidemiology

What It Is, and Who It Is About

Built around peer-reviewed evidence from Zimbabwe and Southern Africa — HIV care, cholera and typhoid outbreaks, and national population surveys.



Introduction to Fundamentals of Epidemiology

Every clinical question you ask — about an HIV cohort in Harare, a cholera outbreak in Manicaland, or hypertension in older patients on ART — is, at its core, an epidemiological question. This session builds the foundations.

By the end of this session, you will be able to:

- 1 Explain what epidemiology is, and understand the concept of population in epidemiology.
- 2 Describe any health-related event using person, place and time.
- 3 Identify the exposure and the outcome in any given study.
- 4 Explain descriptive vs analytical, and observational vs interventional epidemiology.
- 5 Understand the importance of epidemiology to clinical practice in Zimbabwe and Southern Africa.

WHY IT MATTERS

Every other topic in this course — study design, bias, sampling, causation — is built on these foundations.

This topic feeds directly into your weekly quiz and the Final Presentation Project.

Why Epidemiology Matters to a Clinician

Epidemiology is often in the news — an outbreak, a diet linked to disease, a vaccine's effectiveness. Behind almost every such headline sits a piece of epidemiological evidence, good or bad. Learning to read it critically is the purpose of this course.

34,549

Suspected cholera cases reported across all 10 provinces of Zimbabwe during the 2023–2024 outbreak — a number that exists only because someone was counting, in a defined population, over a defined time (WHO AFRO, 2024).

12.9%

HIV prevalence among Zimbabwean adults aged 15–64 in 2020, down from 13.4% in 2015–16 — only possible because epidemiologists tested a defined population at two points in time (ZIMPHIA, Gonese et al., 2020).

Both figures required the same basic machinery: a defined population, a clear case definition, and a count over a defined time. This topic builds that machinery.

What Is Epidemiology?

DEFINITION

"The study of the occurrence and distribution of health-related states or events in specified populations, including the study of determinants influencing such states, and the application of this study to control the health problems."

Porta, Greenland & Last, A Dictionary of Epidemiology, 5th ed., 2008

Four questions sit inside that definition:

● How often?

Frequency — how common is the health event?

● In whom, where, when?

Distribution — who gets it, and at what time and place?

● Why?

Determinants — what drives it?

● So what?

Application — what can we do about it?

A Short History: Comparison Is Fundamental

Epi (upon) + demos (people) — Greek roots, first used in English in 1850. But the defining idea is older than the word: comparing groups who have a disease with groups who do not.

- **WILLIAM FARR (1839)**

Set up England's first routine death-registration system; compared mortality by occupation and place.

- **IGNAZ SEMMELWEIS (1840s)**

Compared maternal mortality between two Vienna clinics; introduced handwashing — an early intervention study.

- **JOHN SNOW (1854)**

Mapped London cholera deaths by water company — descriptive, then analytical epidemiology, without a microscope.

A MODERN PARALLEL — MBARE, HARARE

Polonsky et al. (2014, PLoS One) and Muti, Gombe, Tshimanga et al. (2014, Pan Afr Med J) investigated a 2011–12 typhoid outbreak in Mbare — door-to-door “shoe-leather” case-finding, exactly as Snow practised in 1854.

The Concept of Population in Epidemiology

● Epidemiology

Population is the collection of units from which our sample is drawn — people, institutions, or events. Defined by the research question, not geography.

● Demography

Population means the inhabitants of a country or area — a geographic, administrative concept.

REGIONAL EXAMPLE — THE TENART COHORT, NEWLANDS CLINIC

Shamu et al. (2017, PLoS ONE) defined their population precisely: all patients recruited into care and commenced on ART by 31 December 2006. Not “PLHIV in Zimbabwe” — an exact enrolment list. The denominator is exact, which is what makes every rate calculated from it meaningful.

Describing a Health Event: Person, Place and Time

As epidemiologists, our basic questions about an outcome in a population are “who, where and when?” Three dimensions describe any health event completely.

- **WHO — PERSON**

Age, sex, occupation, HIV status, ART regimen, BMI.

- **WHERE — PLACE**

Country, province, district, clinic catchment, urban vs rural.

- **WHEN — TIME**

Calendar period, secular trend, seasonal pattern.

A PUBLISHED STATEMENT, DECONSTRUCTED

HIV-positive adults aged 50 years and older

Harare, Zimbabwe

February 2004 and March 2020

(after Shamu et al., 2021, PLoS ONE)

SECTION 4 • TWO KINDS OF QUESTION

Descriptive vs Analytical Epidemiology

● Descriptive

“What is happening?” Describes frequency & distribution of an exposure or outcome, without explaining it. Answers who, what, where, when.

● Analytical

“Why is it happening?” Examines associations between exposures and outcomes, to identify possible causes.

SAME COHORT, TWO KINDS OF QUESTION

Descriptive: Shamu et al. (2021, PLoS ONE) — baseline hypertension prevalence among patients ≥ 50 enrolling at Newlands Clinic was 15%.

Analytical: Chimbetete et al. (2017, Trop Med Int Health) — Cox regression tested whether baseline hypertension and CD4 count predicted incident type 2 diabetes among 4,110 patients.

Caution: an association does not by itself prove causation — we return to this later in the module.

SECTION 4 • TWO KINDS OF QUESTION

Exposures and Outcomes

Exposure

Any factor that may influence the outcome — biological, behavioural, environmental, structural.

Outcome

The disease, event or health state of interest — need not be a disease.

Study	Exposure	Outcome
Cigarette smoking & lung cancer	Smoking	Lung cancer
ART regimen & new-onset diabetes (Newlands)	ART regimen / CD4	Incident T2DM
Water source & 2023–24 cholera, Zimbabwe	Water source / sanitation	Cholera case or death

In Snow's 1854 study: water source was the exposure; cholera deaths were the outcome. Area and elevation were merely descriptive.

Observational vs Interventional Epidemiology

● Observational

We describe distribution or determinants without any attempt to influence them. Snow's cholera study was entirely observational.

● Randomised Trial

Participants or whole communities randomly assigned to intervention vs comparison.

● Interventional

We actively modify an exposure and measure its effect. All interventional studies are analytical.

● Quasi-experimental

Intervention introduced, but allocation is not random — e.g. before/after a vaccine rollout.

CLUSTER-RANDOMISED TB TRIAL, HARARE

Corbett et al. (2010, Lancet): 46 neighbourhood clusters randomised to two TB case-finding strategies. Mobile-van strategy found 1.48× more infectious cases than door-to-door visits — interventional, and therefore analytical.

The Role of Epidemiology: Four Functions



ALL FOUR FUNCTIONS, ONE ZIMBABWEAN EXAMPLE — DOLUTEGRAVIR ROLL-OUT

Describe: weight gain documented after switching to dolutegravir-based ART at Newlands Clinic. Interpret: Chimbetete et al. (2017) tested whether weight gain or the drug itself predicted new hypertension/diabetes. Apply: findings informed NCD-screening protocols for PLHIV on dolutegravir across Southern Africa. Evaluate: ongoing cohort follow-up monitors whether those protocols catch cases earlier.

SECTION 7 • POPULATION THINKING

Population vs High-Risk Strategy

Geoffrey Rose distinguished two questions: what determines which individuals get a disease, and what determines which populations get a disease. The distinction shapes the public health response.

● High-risk strategy

Target individuals at greatest risk. Efficient when intervention is costly or risk is concentrated.

● Population strategy

Shift risk distribution across the whole population — often larger overall benefit.

CHOLERA RESPONSE, ZIMBABWE 2023–2024

High-risk: treatment centres & targeted oral cholera vaccine (2.4 million doses) in Harare, Manicaland & Masvingo (>77% of cases).
Population: nationwide WASH improvements and risk communication across all 10 provinces (WHO AFRO, 2024). Neither strategy alone would have worked.

Key Epidemiological Concepts

Concept	Definition
Population	A defined group of individuals under study
Incidence	Number of new cases occurring over a specified period
Prevalence	Total number of existing cases at a point or over a period
Risk factor	A characteristic associated with increased likelihood of disease
Exposure	Contact with a factor that may influence health
Outcome	The health event being measured
Surveillance	Ongoing collection and analysis of health data
Epidemic	Disease occurrence above the expected level in a population
Endemic	Constant presence of a disease within a geographic area or population
Pandemic	An epidemic that spreads across multiple countries or continents

The Fundamentals, Side by Side

Concept	Key question	Regional example
Population	What is the denominator?	TENART cohort enrolment list, Newlands Clinic
Person, place, time	Who, where, when?	Hypertension in HIV+ adults ≥ 50 , Harare, 2004–2020
Descriptive epidemiology	What and how much?	Cholera case counts by province, Zimbabwe 2023–24
Analytical epidemiology	Why?	Cox regression of CD4 & ART regimen vs incident T2DM
Exposure vs outcome	Which factor may influence which?	Water source (exposure) $\rightarrow$ cholera death (outcome)
Observational vs interventional	Did we watch, or did we assign?	DETECTB cluster-randomised TB trial, Harare

Key Takeaways

- ✓ Epidemiology studies the occurrence and distribution of health-related events in specified populations, and applies that to control health problems.
- ✓ The epidemiological population is the defined group from which cases arise — not a geographic concept. Always ask: what is the denominator?
- ✓ Describe any health event using person, place and time.
- ✓ Descriptive epidemiology answers ‘what and how much?’; analytical epidemiology answers ‘why?’.
- ✓ Every study measures an exposure and an outcome — the same variable can be either, depending on the question.
- ✓ Observational studies watch; interventional studies actively assign the exposure. All interventional studies are analytical.
- ✓ The four functions of epidemiology — describe, interpret, apply, evaluate — usually unfold in sequence, as in Zimbabwe’s cholera and HIV programmes.

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